

Power BI Bonus Workshop: Course Attendance Analysis

Description

In this set of exercises, you will start by exploring a flat file in Excel, then import that file into Power BI using Power Query. You will then enhance the data with measures and new calculated columns. Then you will create a report with interactive visualizations, and finally you will share that report in Power BI service.

While this bonus workshop is available starting Week 9, you may begin this workshop at any time during Weeks 9, 10, or 11, and take as much time as you need to work through as much as you can. Some of this skills utilized in this workshop go beyond the Week 9 material, so you may need to do some independent learning as you go.

For the purposes of this exercise, imagine you are an analyst at a company that provides instructor-led training. The company has started offering a new course, called Analytics Academy, and the team that created the course material also built an attendance-tracking spreadsheet to be used by all the instructors who teach the course. You have been tasked with creating a report that the company can provide to the instructional client regarding their employees' attendance.

Getting Started

1. Download **Attendance.zip** from the course material, and unzip to your current week's project folder as **Attendance**. Open this folder and review the three files it contains.
 - * What information is found in this file?
 - * What sets of data are included?
 - * How is it organized?
2. Brainstorm your ideas for the data model to use for this data set. Use a mind-mapping tool like Xmind or draw.io, a whiteboarding app, or a piece of paper, to sketch out your proposed data model.
 - * What fact table(s) do you need to create? (What's the time-series data where things happened?)
 - * What dimension table(s) do you need to create? (What are the pieces of information that describe or expand on information?)
 - * It's also valuable to consider the intended final product. What might you do with this data? What other data might this be combined with to produce useful insights?
3. When you are ready to get started importing and restructuring the data, open a new project in Power BI and save it to your homework folder as **AttendanceAnalyticsAcademy_Step01.pbix**

4. Click on **Get Data** in the **Home** ribbon to launch the Get Data dialog, and choose Excel as source.
5. Start by importing just the first attendance file, **AttendanceAnalyticsAcademy-2023-02-06- EC.xlsx**
 - * Select the Excel sheet named **Sheet1** and click **Transform Data** to import it into Power Query.

1.A.2 Review the data in Power Query, which will look something like this:

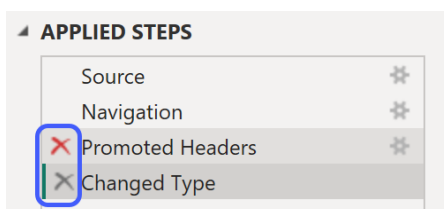
Column1	Course Name	Analytics Academy	Column4	Students: Please fill in "1" for full-day attendance. If you atte...	Column6	Column7
1	Course Instructor	Bess Ryan				
2	Course Start Date	2/6/2023				
3	Course End Date	2/24/2023				
4						
5				Week 1		
6				Day 1	Day 2	Day 3
7	ID	Name	Preferred Name	Email	2/6/2023	2/7/2023
8	1	Michael Scott	Michael	MichaelScott@dundermifflin.com	1	1
9	2	Dwight Schrute	Dwight	DwightSchrute@dundermifflin.com	0.5	1
10	3	Jim Halpert	Jim	JimHalpert@dundermifflin.com	1	1
11	4	Pam Beesly	Pam	PamBeesly@dundermifflin.com	1	1
12	5	Ryan Howard	Ryan	RyanHoward@dundermifflin.com	1	1

You will observe that because of the starting format of the spreadsheet, the data is not neatly organized into a column structure. Consider the different buckets of information to be found in three different areas of the spreadsheet:

Column1	Course Name	Analytics Academy	Column4	Students: Please fill in "1" for full-day attendance. If you atte...	Column6	Column7
1	Course Instructor	Bess Ryan				
2	Course Start Date	2/6/2023				
3	Course End Date	2/24/2023				
4						
5				Week 1		
6				Day 1	Day 2	Day 3
7	ID	Name	Preferred Name	Email	2/6/2023	2/7/2023
8	1	Michael Scott	Michael	MichaelScott@dundermifflin.com	1	1
9	2	Dwight Schrute	Dwight	DwightSchrute@dundermifflin.com	0.5	1
10	3	Jim Halpert	Jim	JimHalpert@dundermifflin.com	1	1
11	4	Pam Beesly	Pam	PamBeesly@dundermifflin.com	1	1
12	5	Ryan Howard	Ryan	RyanHoward@dundermifflin.com	1	1

You will need to restructure this data by separating these buckets into useful tables.

6. Start by right clicking on Sheet1 and renaming it SourceRaw.
7. In the **Applied Steps** panel on the right, delete the steps that Power Query automatically added for **Promoted Headers** and **Changed Type** by clicking the icon to the left of the step name. (Delete each step from the bottom first to avoid warning messages about affecting subsequent steps.)



8. Right click the renamed SourceRaw and select **Reference**. This will create a new query that points back to the original. As you work, any changes made to the original query will also be reflected in the referencing query. This is useful when creating multiple queries that use the same data source and same set of transformations.
9. Rename the new query as Course. This will become your table for course-level information, which is currently contained in the first four rows, in Column 2 and Column 3.
10. Viewing Course, in the **Home** tab of the ribbon, select **Keep Rows**, then **Keep Top Rows**. Enter the number of rows to keep, 4, and click OK.

The screenshot shows the Power Query Editor interface. The ribbon is set to 'Home'. The 'Keep Rows' button is highlighted in the 'Reduce Rows' group. A dialog box titled 'Keep Top Rows' is open, asking 'Specify how many rows to keep.' with a text box containing the number '4'. The background shows a table with columns 'Column1', 'Column2', 'Column3', and 'Column4'. The first four rows of data are visible, corresponding to the 'Course' query.

Column1	Column2	Column3	Column4
1	Course Name	Analytics Academy	
2	Course Instructor	Elise Christenson	
3	Course Start Date	2/6/2023	
4	Course End Date	2/24/2023	

11. Now remove unnecessary columns.

- * In the **Home** tab, select **Choose Columns**. Use the dialog to un-check all columns, then check the boxes only for **Column2** and **Column3**.
- * Alternatively, you could use Ctrl+click to select these columns in the editor, then right click the column header and choose **Remove Other Columns** from the menu.

12. This leaves two columns with four rows of data:

	Column2	Column3
1	Course Name	Analytics Academy
2	Course Instructor	Elise Christenson
3	Course Start Date	2/6/2023
4	Course End Date	2/24/2023

13. Right now, the information categories are all organized in one column, and specific data points are lumped together in the next column.

- * To rearrange the data from vertical to horizontal so that the data is organized into one column for each category of information, in the ribbon click on the **Transform** tab, and click **Transpose**.
- * In the **Transform** tab, click **Use First Row as Headers** to promote first row to the header.

14. Now you have a table including basic info about this particular course, including the name, instructor, start date, and end date. But notice that there is no unique identifier for this course. What if later you want to aggregate this information with other sections of the course taught by other instructors or on different dates? To solve this problem, add a column that creates a unique key for the course.

- * Use the **Add Column** tab in the ribbon to create a **Custom Column**. Name the column **CourseID**.
- * In the column formula, include the **Course Name**, **Course Instructor**, and **Course Start Date**, concatenated with & and including a | separator between each. Note that because the **Course Start Date** has a data type of **date**, you will need to use the **Text.From** function to add it as text in the new column. In the custom column dialog, your entered values should look like this:

```
[Course Name] & " " & "|" & " " & [Course Instructor] & " " & "|" & " " & Text.From([Course Start Date])
```

- * Click OK to create the new column. The result should look like this:

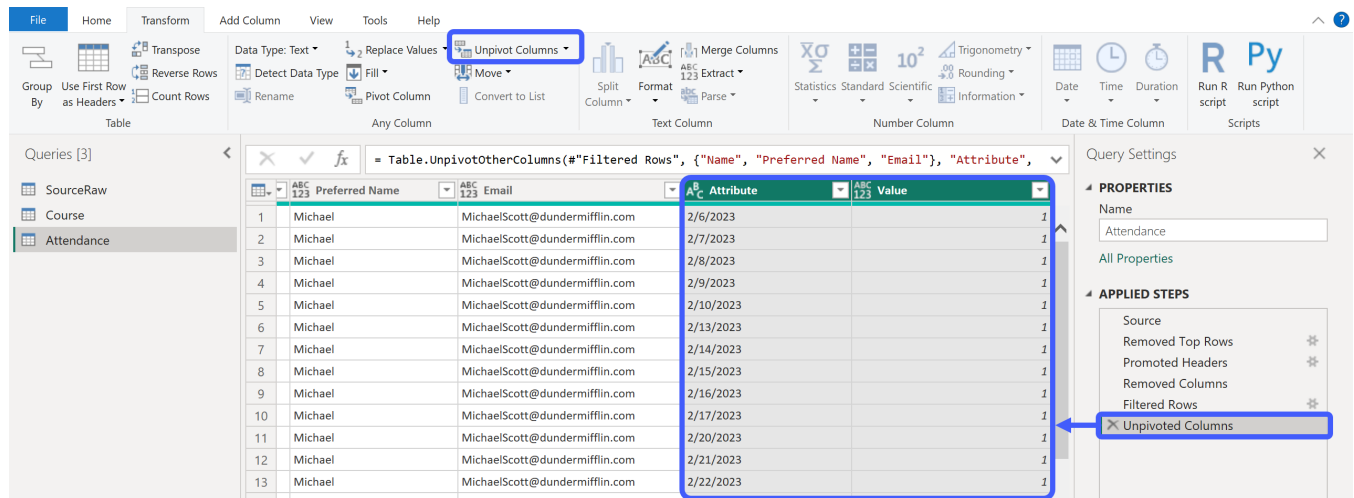
	Course Instructor	Course Start Date	Course End Date	CourseID
1	Elise Christenson	2/6/2023	2/24/2023	Analytics Academy Elise Christenson 2/6/2023

15. Save your file as **AttendanceAnalyticsAcademy_Step01.pbix**

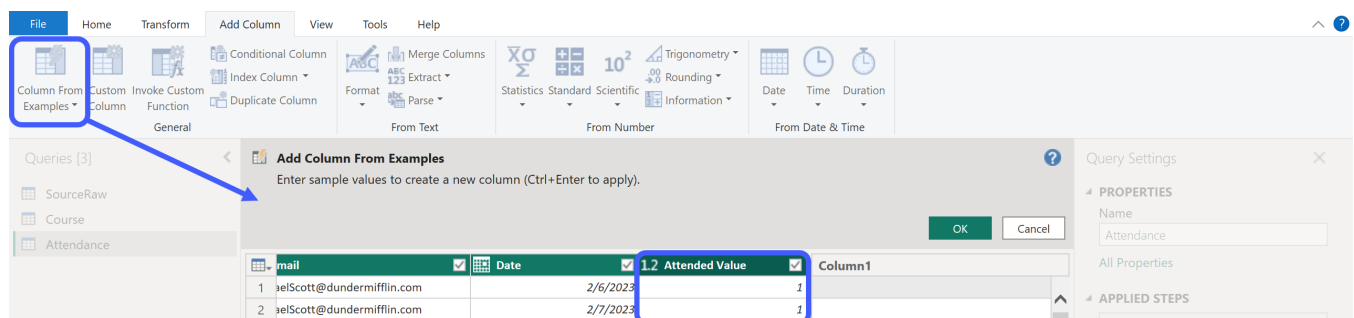
Transformation Part 2: Breaking Out Attendance Data

1. In the queries, right click **SourceRaw** and again select **Reference**. Rename the new query **Attendance**. This will be your central fact table.
2. The top rows do not contain any information relevant to daily attendance and can be removed. In the **Home** tab, use **Remove Rows > Remove Top Rows** to delete the top 7 rows from **Attendance**.
3. In the ribbon, click **Use First Row as Headers**.
4. Observe that the first column contains an **ID** which is a row number for the attendance sheet. You will not be using this information, so delete this column.

5. Notice that the bottom rows do not contain any data, only null values. To remove these null records, click the drop-down button  in the header in any column, and then on **Remove Empty** (or alternatively, uncheck **(null)**) to filter out the null rows.
6. At this point you have a row for each student, with columns for each date of the course and the value representing that student's attendance on that date. To aggregate these dates and values into a single column for each, you will need to use **unpivot**.
 - * Select all the **date columns** by clicking the header to select the first date column, **2/6/2023**, then holding shift and using the right arrow key on your keyboard (or End key) to scroll to the far right, selecting all through the last column, **2/24/2023**.
 - * Then in the **Transform** tab, click **Unpivot Columns**. The result should look like this:



7. Rename the new columns: Change **Attribute** to **Date**, and change **Value** to **Attended Value**.
8. Check the data types for your columns. Update the **Date** column to the **Date** data type, and **Attended Value** to the **Decimal Number (1.2)** data type.
9. Next add a column to represent attendance with words instead of numbers.
 - * In the **Add Column** tab, click **Column From Examples**.
 - * Scroll as needed so that you can see the **Attended Value** column in the “Add Column from Examples” wizard.



- * On the right, double click on the new **Column1** and change the new column name to **Attendance Type**.
- * Begin typing in a few sample values in the rows:
 - **Attended** where the **Attended Value** is 1
 - **Partially Attended** where the **Attended Value** is 0.5
 - **Absent** where the **Attended Value** is 0 (you may have to scroll down a ways to find an example!)
- * As you add a couple samples for each value of **Attended Value**, you should see the gray suggested values match up to the intended scheme.
- * When you are satisfied with the new column values, click OK at the top to create the column.

Hint: at the top, you should see the transformation logic appear as follows:

Transform: if [Attended Value] = 1 then "Attended" else if [Attended Value] = 0.5 then "Partially Attended" else if [Attended Value] = 0 then "Absent" else null

(Depending on the width of your screen the full logic may be cut off in the preview, as in the screenshot below.)


Add Column From Examples
?

Enter sample values to create a new column (Ctrl+Enter to apply).
 Transform: if [Attended Value] = 1 then "Attended" else if [Attended Value] = 0.5 then "Partially Attended" else if [Attended Value] = 0 then...

	mail	Date	1.2 Attended Value	Attendance Type
1	aelScott@dundermifflin.com	2/6/2023	1	Attended
2	aelScott@dundermifflin.com	2/7/2023	1	Attended
3	aelScott@dundermifflin.com	2/8/2023	1	Attended
4	aelScott@dundermifflin.com	2/9/2023	1	Attended
5	aelScott@dundermifflin.com	2/10/2023	1	Attended
6	aelScott@dundermifflin.com	2/13/2023	1	Attended
7	aelScott@dundermifflin.com	2/14/2023	1	Attended
8	aelScott@dundermifflin.com	2/15/2023	1	Attended
9	aelScott@dundermifflin.com	2/16/2023	1	Attended
10	aelScott@dundermifflin.com	2/17/2023	1	Attended
11	aelScott@dundermifflin.com	2/20/2023	1	Attended
12	aelScott@dundermifflin.com	2/21/2023	1	Attended
13	aelScott@dundermifflin.com	2/22/2023	1	Attended
14	aelScott@dundermifflin.com	2/23/2023	1	Attended
15	aelScott@dundermifflin.com	2/24/2023	1	Attended
16	ntSchrute@dundermifflin.com	2/6/2023	0.5	Partially Attended

10. Save your file as **AttendanceAnalyticsAcademy_Step02.pbix**

Transformation Part 3: Adding Dimension Tables

- Currently, all the student data is still folded into the **Attendance** table. Normalize this by creating a dimension table that breaks out student information:
 - Right click on Attendance, and choose **Duplicate**.
 - Rename the new query as **Students**.
- Go back to **Attendance**. Clean up the query by deleting the columns for **Name** and **Preferred Name** (but leave the column **Email**).
- In **Students**, remove columns that do not contain student information (**Date**, **Attended Value**, and **Attendance Type**).
- What do you think of the remaining column names? Think about how these are likely to be used in visualizations.
 - "Name" is a little bit ambiguous since this column contains both first name and last name.
 - "Preferred Name" is ok but a little clunky given how we might be likely to use this field to label information in our visualizations.
 - Instead, rename the original Name column to **Full Name**, then rename Preferred Name to just **Name**.
- Right click on the column header for **Email** and select **Remove Duplicates**.

The screenshot shows a data table with the following columns: Full Name, Name, and Email. The data consists of 15 rows, all representing 'Michael Scott' with the email 'MichaelScott@dunde'. A right-click context menu is open over the 'Email' column header, and the 'Remove Duplicates' option is highlighted. The background interface shows a list of queries on the left, with 'Students' selected. On the right, the 'Query Settings' panel shows the 'Name' property set to 'Students' and a list of 'APPLIED STEPS' including 'Removed Top Rows', 'Promoted Headers', 'Removed Columns', 'Filtered Rows', 'Unpivoted Columns', 'Renamed Columns', 'Changed Type', 'Added Conditional Column', 'Removed Columns1', and 'Renamed Columns1'.

Your de-duplicated list should include 20 students.

- Save your file as **AttendanceAnalyticsAcademy_Step03.pbix**
- Next, to create a date dimension table, right click once again on **SourceRaw** and select **Reference**. Rename the new query **Course Day**.

8. For this table, we are interested in the week- and day-level information that had formed the headers of the attendance section of the spreadsheet. None of the information in the first 4 columns is needed for Course Day, so these should be removed.
9. Notice that information on the starting week of the course begins at row 6. Use **Remove Rows** to delete the top 5 rows which are not needed.
10. For this date table, you only want to keep the rows for week, day, and date. Use **Keep Rows** to keep only these 3 top rows and remove other rows.

Queries [5]

- SourceRaw
- Course
- Attendance
- Students
- Course Day

Query Settings

PROPERTIES

Name

Course Day

APPLIED STEPS

- Source
- Removed Columns
- Removed Top Rows
- Kept First Rows

11. In the **Transform** tab of the ribbon, use **Transpose** to convert the rows to columns. Rename the columns **Week Of Course**, **Day Of Course**, and **Full Date**.

Queries [5]

- SourceRaw
- Course
- Attendance
- Students
- Course Day

Query Settings

PROPERTIES

Name

Course Day

APPLIED STEPS

- Source
- Removed Columns
- Removed Top Rows
- Kept First Rows
- Transposed Table
- Renamed Columns

12. For the new column **Full Date**, change the data type to a Date type.
13. Select the **Week of Course** column. Still in the **Transform** tab, use **Fill > Down** to populate the null values based on existing data.

Queries [5]

- SourceRaw
- Course
- Attendance
- Students
- Course Day

Query Settings

PROPERTIES

Name

Course Day

APPLIED STEPS

- Source
- Removed Columns
- Removed Top Rows
- Kept First Rows
- Transposed Table
- Renamed Columns
- Fill Down

14. In the **Add Column** tab, use **Column from Examples** to add a new column called **Date**. Begin entering example values starting with 2 / 6 (without the year), and confirm that the gray suggested values update to conform to this pattern. Click OK to create the column.

Add Column From Examples

Enter sample values to create a new column (Ctrl+Enter to apply).
 Transform: `Text.BeforeDelimiter(Text.From([Full Date], "en-US"), "/", 1)`

OK Cancel

ABC 123	Week of Course	ABC 123	Day of Course	ABC 123	Full Date	Date
1	Week 1		Day 1		2/6/2023	2/6
2	Week 1		Day 2		2/7/2023	2/7
3	Week 1		Day 3		2/8/2023	2/8
4	Week 1		Day 4		2/9/2023	2/9
5	Week 1		Day 5		2/10/2023	2/10

15. Use **Column from Examples** twice more to add new columns for **Week Of Course Number** (based on Week of Course) and **Day Of Course Number** (based on Day of Course), so that your new columns match the below screenshot:

Query Settings

Query: `= Table.AddColumn("#Inserted Text After Delimiter", "Day of Course Number", each`

Irse	ABC 123	Full Date	A ^B C	Date	A ^B C	Week of Course Number	A ^B C	Day of Course Number
1		2/6/2023	2/6		1		1	
2		2/7/2023	2/7		1		2	
3		2/8/2023	2/8		1		3	
4		2/9/2023	2/9		1		4	
5		2/10/2023	2/10		1		5	
6		2/13/2023	2/13		2		6	
7		2/14/2023	2/14		2		7	
8		2/15/2023	2/15		2		8	
9		2/16/2023	2/16		2		9	
10		2/17/2023	2/17		2		10	
11		2/20/2023	2/20		3		11	
12		2/21/2023	2/21		3		12	
13		2/22/2023	2/22		3		13	
14		2/23/2023	2/23		3		14	
15		2/24/2023	2/24		3		15	

Query Settings

PROPERTIES

Name: Course Day

APPLIED STEPS

- Source
- Removed Columns
- Removed Top Rows
- Kept First Rows
- Transposed Table
- Renamed Columns
- Filled Down
- Inserted Text Before Delimiter
- Inserted Text After Delimiter
- Inserted Text After Delimiter1

16. Click on the **Full Date** column to select it. Still on the **Add Column** tab of the ribbon, use the **Date** options to add a new column for **Year**.

File Home Transform Add Column View Tools Help

Column From Examples Custom Column Invoke Custom Function Duplicate Column

Conditional Column Index Column

Format Merge Columns

Statistics Standard Scientific Rounding Information

From Text From Number

Queries [5]

- SourceRaw
- Course
- Attendance
- Students
- Course Day

Query: `= Table.AddColumn("#Inserted Text After Delimiter", "Date`

Irse	Full Date	A ^B C	Date	A ^B C	Week
1	2/6/2023	2/6		1	
2	2/7/2023	2/7		1	
3	2/8/2023	2/8		1	
4	2/9/2023	2/9		1	
5	2/10/2023	2/10		1	
6	2/13/2023	2/13		2	
7	2/14/2023	2/14		2	
8	2/15/2023	2/15		2	
9	2/16/2023	2/16		2	

Date

- Age
- Date Only
- Parse
- Year
- Month
- Quarter
- Week
- Day
- Subtract Days
- Combine Date and Time
- Earliest
- Latest

Query Settings

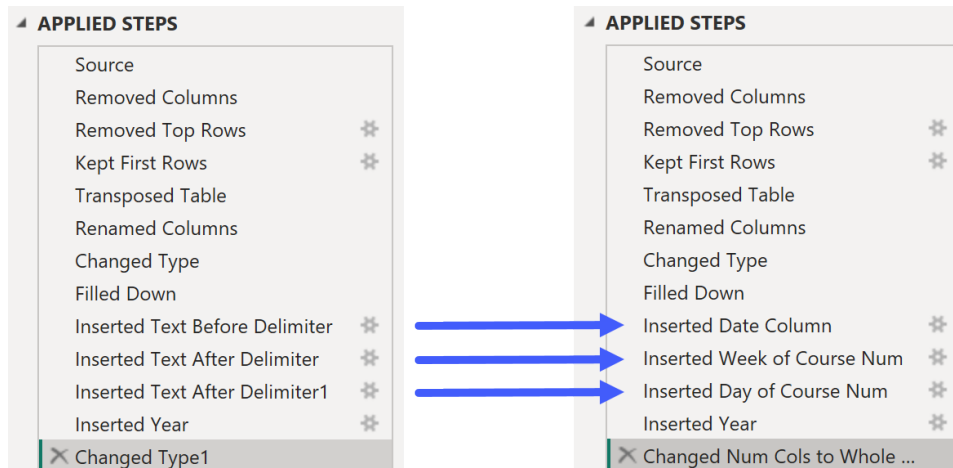
PROPERTIES

Name: Course Day

APPLIED STEPS

- Source
- Removed Columns
- Removed Top Rows
- Kept First Rows
- Transposed Table

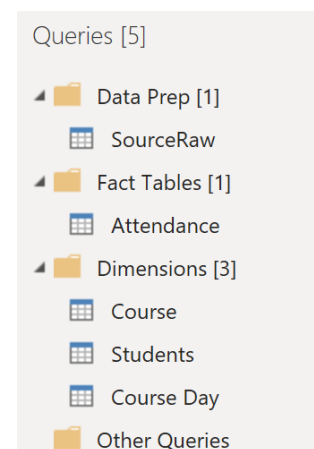
17. Check the data types for your columns, and update **Week of Course Number** and **Day of Course Number** to the data type **Whole Number**.
18. Take a quick look at the list of Applied Steps. At this point, you will have several steps labeled **Inserted Text Before/After Delimiter**. To make it easier to retrace your steps, you can right-click the name of the step and choose **Rename**, then change it to a more descriptive label such as “Inserted Date Column” or “Inserted Week Of Course Number.”



Note: if you applied changes in a slightly different order from the instructions, that's ok! If you relabel your steps, it is more important to match *what you actually did* rather than exactly matching the screenshot above.

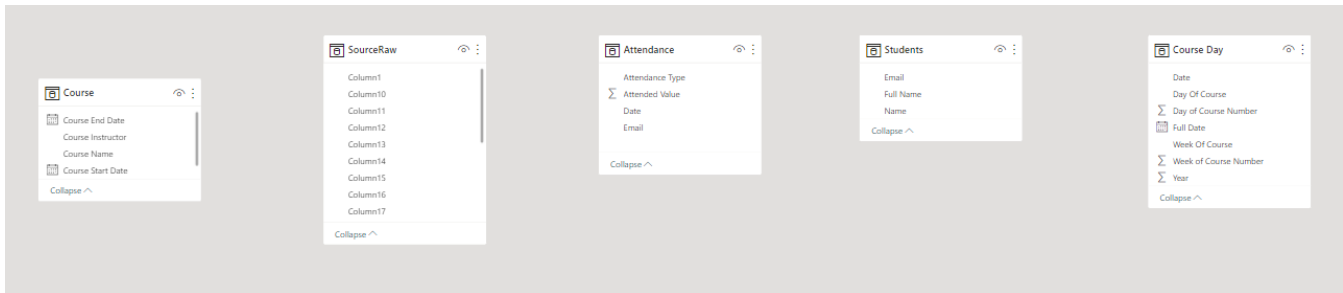
19. You might also consider grouping our tables to help distinguish preparatory steps from final fact / data tables and dimension / lookup tables. While not strictly necessary, especially when dealing with a small number of tables such as you have here, this can be helpful when managing many tables or simply to help make your intentions clear for another user (or your future self).
- * In the left pane, right click in the gray background and select **New Group**. Name the group **Data Prep**, then click OK. Repeat steps to create two more groups, **Fact Tables** and **Dimensions**.
 - * Click and drag **SourceRaw** into **Data Prep**.
 - * Click and drag **Course** into **Dimensions**. Do the same with **Students** and **Course Day**.
 - * Click and drag **Attendance** into **Fact Tables**.
 - * When you are done, your groups should look like this →

20. Save your file as **AttendanceAnalyticsAcademy_Step04.pbix**

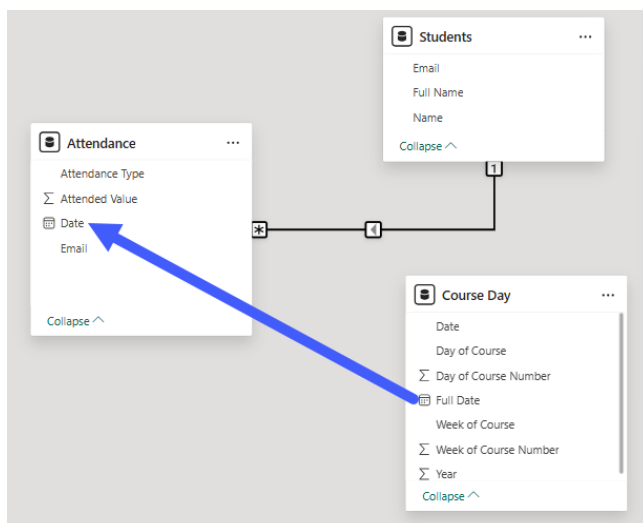


Modeling the Attendance Data

1. Click **Close & Apply** to return to Power BI Desktop.
2. Go to the **Model view**.
3. Notice that your tables are present but not in any particular order. You may have one relationship detected, or Power BI may not have detected any relationships.



4. Move **SourceRaw** out of the way to a lower corner, and click the eye icon to hide it from display. You will not be using this table in report creation.
 5. Move the fact table, **Attendance**, to the center of the canvas.
 6. Arrange the remaining dimension tables around the fact table.
 7. Create relationships between the fact table and each dimension table by dragging the appropriate “foreign key” from the fact table and dropping it onto the corresponding “primary key” in the dimension table (or vice versa).
- * **Attendance** and **Course Day** can be connected by dragging the **Full Date** field in **Course Day** onto the **Date** field in **Attendance**.



In the **New relationship** pop-up window, make sure that the correct column names are selected for each table. The cardinality should be detected as one-to-many.

New relationship

Select tables and columns that are related.

From table
Course Day

Date	Day of Course	Day of Course...	Full Date	Week of Cour...	Week of Cour...	Year
2/6	Day 1	1	Monday, Febr...	Week 1	1	2023
2/7	Day 2	2	Tuesday, Febr...	Week 1	1	2023
2/8	Day 3	3	Wednesday, F...	Week 1	1	2023

To table
Attendance

Attendance T...	Attended Value	Date	Email
Attended	1	Monday, Febr...	MichaelScott...
Attended	1	Tuesday, Febr...	MichaelScott...
Attended	1	Wednesday, F...	MichaelScott...

Cardinality
One to many (1:*)

Cross-filter direction
Single

☒ Make this relationship active
☐ Assume referential integrity

☐ Apply security filter in both directions

Save **Cancel**

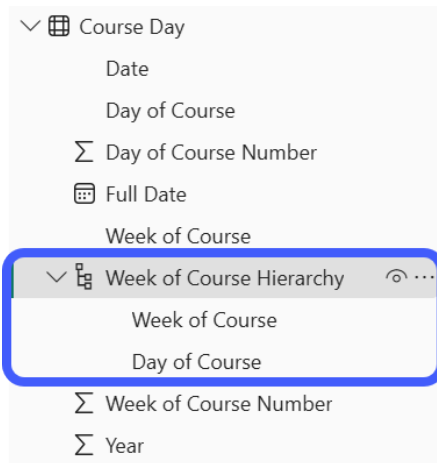
- * If the relationship is not automatically created between **Attendance** and **Students**, these can be connected on the **Email** field, which acts as a foreign key in **Attendance** and a unique primary key in **Students**.
 - * How can you connect **Attendance** and **Course**? Any issues here?
Hint: with only the starting data, it *seems* like you could make a relationship between these tables based on the **Course Start Date** in **Course** and the **Date** in **Attendance**. However, if the data is expanded to include multiple courses that have the same start date, then this relationship will turn into a many-to-many relationship, which is not desirable.
8. To create a one-to-many relationship between **Course** and **Attendance**, you will need to add a **CourseID** column in **Attendance** so that each course has a unique identifier.
- * In the ribbon, click **Transform Data** to return to Power Query.
 - * In the **Attendance** table, add a **Custom Column**. Name the column **CourseID**. Since we have only a single value to add for this column, enter the following expression using a hard-coded string under **Custom column formula**, then click OK:

```
= "Analytics Academy | Elise Christenson | 2/6/2023"
```

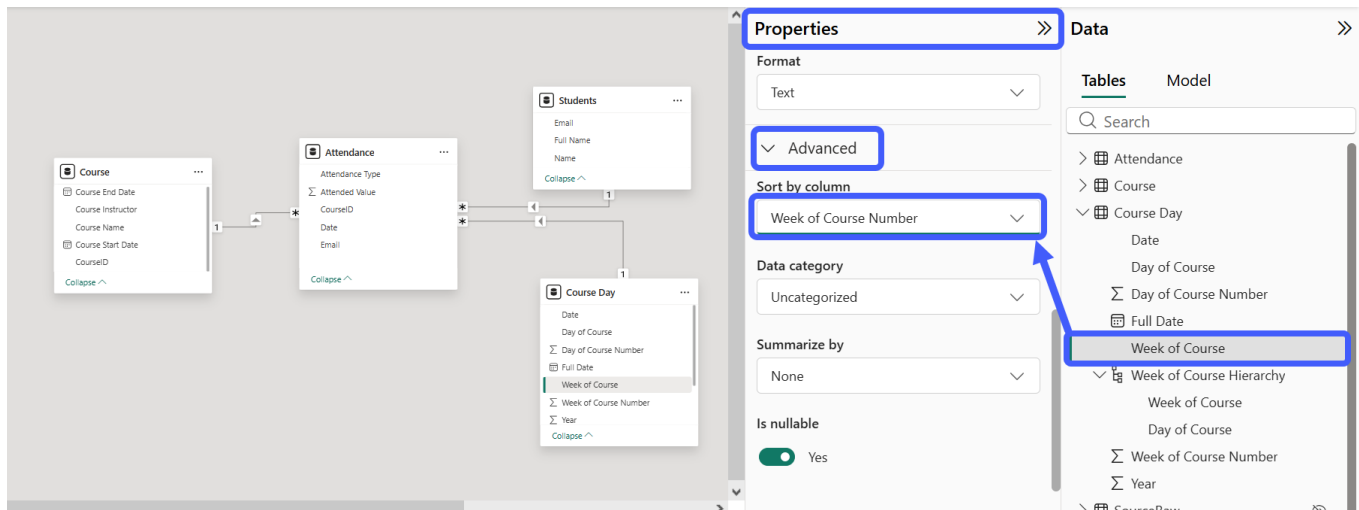
* Once the column is created, click **Close & Apply** to return to Power BI Desktop.

9. Add the relationship between the **Attendance** table and the **Course** table based on **CourseID**, if it's not automatically detected.

10. In the **Course Day** table, add a hierarchy based on **Week Of Course** that includes the levels **Week Of Course** and **Day Of Course**. (Note: you can do this from the Data panel in the Model view.)



11. Select each of the below columns one at a time, and open the Properties panel in the Model view. Under Properties, scroll down to the bottom and expand Advanced. Then update the **Sort by column** for the following columns in Course Day:



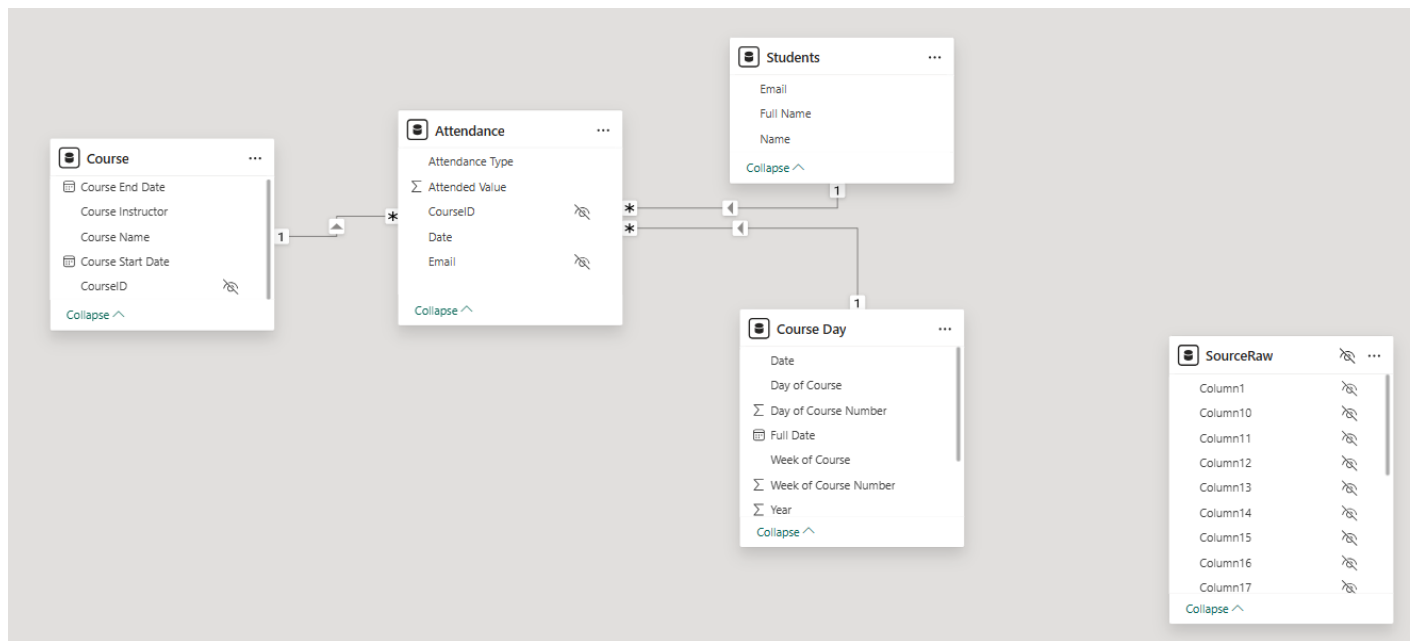
* **Week Of Course** → change Sort by column to **Week of Course Number**

* **Day Of Course** → change Sort by column to **Day of Course Number**

* **Date** → change Sort by column to **Full Date**

12. Review the columns for each table, and hide any columns that will not be used for display in your report output, using the eye icon (either within the table cards or in the Data pane at the right).

13. Your updated data model should look something like this:



14. Save your file as **AttendanceAnalyticsAcademy_Step05.pbix**

Enhancing the Model with DAX

1. Go to the **Table view**. Looking at the **Attendance** table, in the ribbon under **Table Tools** click **New measure**.
2. Create a measure to add up the total number of days actually attended by all students, using the **SUM** function to add up attendance days based on the **Attended Value** column. Call this measure **Total Days Attended**.

* Your DAX formula should end up looking like this:

```
1 Total Days Attended = SUM(Attendance[Attended Value])
```

3. Create another measure to tally the total number of absences. Call this one **Total Absences**. In this calculation, you only want to count the 0 values on the **Attended Value** column. To do so, you will need the iterator function **COUNTAX** to check each value row by row, and the **FILTER** function to narrow down the row values to be counted.

* Use the following expression:

```
Total Absences = COUNTAX( FILTER (Attendance, Attendance[Attended Value] = 0), Attendance[Attendance Type] )
```

(To add line breaks like in the screenshot below, hit shift+enter on your keyboard)

```
1 Total Absences =
2     COUNTAX(
3         FILTER (
4             Attendance,
5             Attendance[Attended Value] = 0
6         ),
7         Attendance[Attendance Type]
8     )
```

4. In order to pass the course, a student must have attended at least 80% of the course days. Create a series of measures to help you create “pass” / “did not pass” visualizations:

* Attendance Goal = COUNTROWS('Course Day')

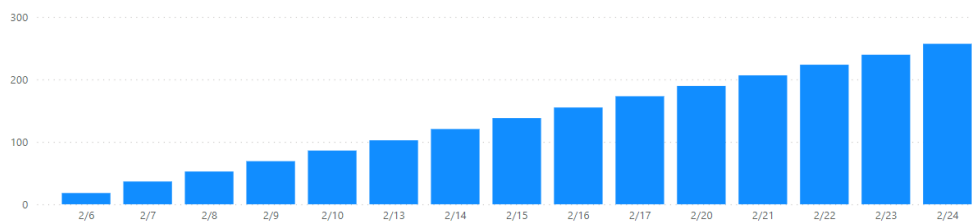
```
* Percent Attendance = DIVIDE(
    SUM(Attendance[Attended Value])
    , Attendance[Attendance Goal]
)
```

```
* Passed Course = IF(
    Attendance[Percent Attendance] >= .8, "Pass", "Did Not Pass"
)
```

5. Here's a trickier one: How would you create a measure that produces a rolling total of days attended by all students?

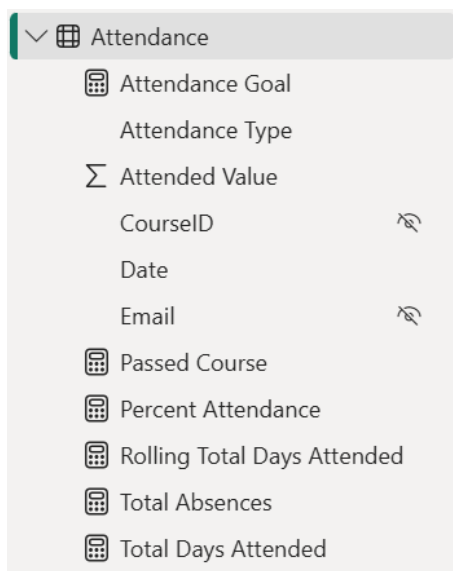
```
* Rolling Total Days Attended =
CALCULATE (
    SUM(Attendance[Attended Value]),
    FILTER (
        ALL(Attendance),
        Attendance[Date] <= MAX (Attendance[Date])
    )
)
```

- * A rolling total will always yield a result that looks something like this because it is adding up the cumulative total of, e.g., the current day plus everything that came before the current day:



- * What is happening at each of the layers in this measure? There's an **ALL** function nested inside a **FILTER** which is nested inside **CALCULATE**. You may need to do a little research to understand how this one works, but it is well worth the payoff! This type of complex DAX calculation can be extremely helpful when building dashboards in Power BI.

6. When you are done with the above, your Attendance table should have 6 new measures in addition to the original columns:



7. Save your file as **AttendanceAnalyticsAcademy_Step06.pbix**

Building the Attendance Report

- Now that you have some robust measures built, it's time to start working on the visualizations for this report. Again keep in mind our scenario: this is a report you are creating for an instructional company to provide to a business client, showing metrics on their employees' attendance of a course.

Imagine that the client manager also sends you an example copy of what the client used to receive from a former training provider:

eTechTeach

Technology Training for the Next Century And Beyond!

Client: ABC Group Course: Q254-Off365Global Course Name: Office 365 for Global Bu... Instructor: Smith Dates: 12-07-2021 30-07-2021									Total Learning Hours: 660 (110 Days) Course Completion Rate: 3 (27%) Eligible for Completion Booster Course: 5 (45%)								
Registered: Schrute,Dwig Halpert,Jim Howard,Ryan Hudson,Stanl Malone,Kevin Palmer,Mered Martin,Angel Lappin-Vance Levinson,Jan Kapoor,Kelly Bratton,Cree					Completed: Halpert,Jim Martin,Angel Bratton,Cree					Partially Completed: Schrute,Dwig Hudson,Stanl Malone,Kevin Palmer,Mered Lappin-Vance							

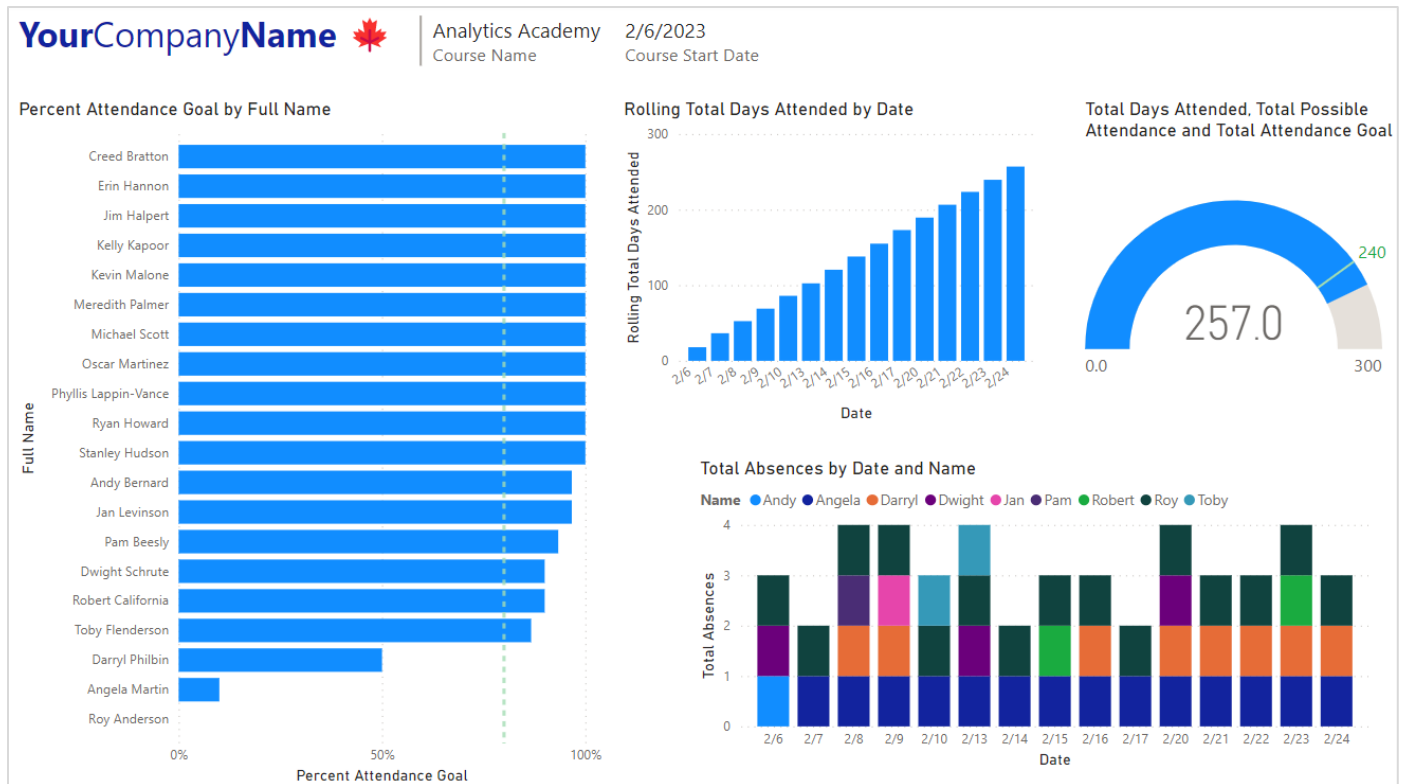
Name^ Day>	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Student Total	Student %
Schrute,Dwig	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	12	80%
Halpert,Jim	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	15	100%
Howard,Ryan	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	2	13%
Hudson,Stanl	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	12	80%
Malone,Kevin	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	12	80%
Palmer,Mered	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	12	80%
Martin,Angel	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	15	100%
Lappin-Vance	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	12	80%
Levinson,Jan	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0%
Kapoor,Kelly	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	3	20%
Bratton,Cree	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	15	100%
Total by Day	10	10	9	8	8	8	8	8	8	8	8	8	3	3	3		
% by Day	91%	91%	82%	73%	73%	73%	73%	73%	73%	73%	73%	73%	27%	27%	27%		

Along with the example, there's a note:

ABC Group really liked having a quick summary of course completion and top level metrics for the course. They also liked having a record of the detailed attendance breakdown, which their HR team mostly used to follow up on excused versus unexcused absences, and to identify who had a 100%

completion rate once they accounted for excused absences. They'd like to get a similar report for our courses ... but "hopefully easier to read!"

2. In a blank document or on scratch paper, jot down a few quick bullet points to answer the following brainstorming questions:
 - * What is the purpose of the report you are creating? What basic information does it need to include? What insights?
 - * From the perspective of your company who is providing the report, what do you want to communicate to the client?
 - * From the perspective of the client, what do they want to know?
3. In Power BI Desktop, click over to the **Report view**.
4. Let's start with some basic information. Create report with two pages that include:
 - * Attendance Summary
 - ◇ A company name and logo, and basic course information
 - ◇ The total course days attended by all students out of a possible total of 300 days and a goal reflecting 80% total attendance
 - ◇ The rolling total of course days attended by all students
 - ◇ The percentage of course days each attended by each student
 - ◇ Total absences by day and by student
 - * Full Attendance List
 - ◇ A matrix of daily attendance by student and by day, color coded to highlight absences and partial absences
 - ◇ A visual depicting course attendance by day
 - ◇ A list of students who passed and did not pass
 - ◇ A list of total absences per student, including partial absences
5. Need some ideas? Check out the the following screenshots for inspiration, but feel free to come up with your own variations:



YourCompanyName



Analytics Academy

2/6/2023

Course Name

Course Start Date

Week Of Course	Week 1										Total	Week 2										Total
	Full Name	Day 1	Day 2	Day 3	Day 4	Day 5	Total	Day 6	Day 7	Day 8		Day 9	Day 10	Total	Day 11	Day 12	Day 13	Day 14	Day 15	Total		
Andy Bernard		0.50	1.00	1.00	1.00	1.00	4.50	1.00	1.00	1.00	1.00	1.00	5.00	1.00	1.00	1.00	1.00	1.00	5.00	14.50		
Angela Martin		1.00	0.50	0.00	0.00	0.00	1.50	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.50		
Creed Bratton		1.00	1.00	1.00	1.00	1.00	5.00	1.00	1.00	1.00	1.00	1.00	5.00	1.00	1.00	1.00	1.00	1.00	5.00	15.00		
Darryl Philbin		1.00	1.00	0.00	0.00	1.00	3.00	1.00	1.00	1.00	0.00	1.00	4.00	0.00	0.00	0.00	0.00	0.50	0.50	7.50		
Dwight Schrute		0.50	1.00	1.00	1.00	1.00	4.50	0.50	1.00	1.00	1.00	1.00	4.50	0.50	1.00	1.00	1.00	1.00	4.50	13.50		
Erin Hannon		1.00	1.00	1.00	1.00	1.00	5.00	1.00	1.00	1.00	1.00	1.00	5.00	1.00	1.00	1.00	1.00	1.00	5.00	15.00		
Jan Levinson		1.00	1.00	1.00	0.50	1.00	4.50	1.00	1.00	1.00	1.00	1.00	5.00	1.00	1.00	1.00	1.00	1.00	5.00	14.50		
Jim Halpert		1.00	1.00	1.00	1.00	1.00	5.00	1.00	1.00	1.00	1.00	1.00	5.00	1.00	1.00	1.00	1.00	1.00	5.00	15.00		
Kelly Kapoor		1.00	1.00	1.00	1.00	1.00	5.00	1.00	1.00	1.00	1.00	1.00	5.00	1.00	1.00	1.00	1.00	1.00	5.00	15.00		
Kevin Malone		1.00	1.00	1.00	1.00	1.00	5.00	1.00	1.00	1.00	1.00	1.00	5.00	1.00	1.00	1.00	1.00	1.00	5.00	15.00		
Meredith Palmer		1.00	1.00	1.00	1.00	1.00	5.00	1.00	1.00	1.00	1.00	1.00	5.00	1.00	1.00	1.00	1.00	1.00	5.00	15.00		
Michael Scott		1.00	1.00	1.00	1.00	1.00	5.00	1.00	1.00	1.00	1.00	1.00	5.00	1.00	1.00	1.00	1.00	1.00	5.00	15.00		
Oscar Martinez		1.00	1.00	1.00	1.00	1.00	5.00	1.00	1.00	1.00	1.00	1.00	5.00	1.00	1.00	1.00	1.00	1.00	5.00	15.00		
Pam Beesly		1.00	1.00	0.00	1.00	1.00	4.00	1.00	1.00	1.00	1.00	1.00	5.00	1.00	1.00	1.00	1.00	1.00	5.00	14.00		
Phyllis Lappin-Vance		1.00	1.00	1.00	1.00	1.00	5.00	1.00	1.00	1.00	1.00	1.00	5.00	1.00	1.00	1.00	1.00	1.00	5.00	15.00		
Robert California		1.00	1.00	1.00	1.00	1.00	5.00	1.00	1.00	0.50	1.00	1.00	4.50	1.00	1.00	1.00	0.00	1.00	4.00	13.50		
Roy Anderson		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00		
Ryan Howard		1.00	1.00	1.00	1.00	1.00	5.00	1.00	1.00	1.00	1.00	1.00	5.00	1.00	1.00	1.00	1.00	1.00	5.00	15.00		
Stanley Hudson		1.00	1.00	1.00	1.00	1.00	5.00	1.00	1.00	1.00	1.00	1.00	5.00	1.00	1.00	1.00	1.00	1.00	5.00	15.00		
Toby Flenderson		1.00	1.00	1.00	1.00	0.00	4.00	0.00	1.00	1.00	1.00	1.00	4.00	1.00	1.00	1.00	1.00	1.00	5.00	13.00		
Total		18.00	18.50	16.00	16.50	17.00	86.00	16.50	18.00	17.50	17.00	18.00	87.00	16.50	17.00	17.00	16.00	17.50	84.00	257.00		

Count of Attendance Type by Date and Attendance Type

Attendance Type

Absent

Attended

Partially Attended

Count of Attend...

2/6

2/7

2/8

2/9

2/10

2/13

2/14

2/15

2/16

2/17

2/20

2/21

2/22

2/23

2/24

Name

Passed Course

Andy

Pass

Creed

Pass

Dwight

Pass

Erin

Pass

Jan

Pass

Jim

Pass

Kelly

Pass

Kevin

Pass

Meredith

Pass

Michael

Pass

Oscar

Pass

Pam

Pass

Phyllis

Pass

Robert

Pass

Ryan

Pass

Stanley

Pass

Toby

Pass

Angela

Did Not Pass

Darryl

Did Not Pass

Roy

Did Not Pass

Total

Pass

Name

Total Absences

Roy

15

Angela

14

Darryl

8

Dwight

3

Robert

2

Toby

2

Andy

1

Jan

1

Pam

1

Total

47

6. Save your file as AttendanceAnalyticsAcademy_Step07.pbix

EXTRA CHALLENGE: Combining & Transforming Multiple Files

1. How do you approach analysis of multiple spreadsheets where complex transformations are required? That is the subject of this set of challenge exercises.
2. Create a new Power BI project and save it as **MultiAttendance_Step01.pbix**
3. As an initial data source, again start with the first the Excel file from the Attendance folder. This time, rename this initial query as **AttendanceExample**. Use the steps that you followed in the first part of the workshop to recreate the previous Attendance table (skipping creating as a reference from SourceRaw, as we are not using that method in this step).
 - * Note that because you will ultimately be pulling together data from multiple spreadsheets, you will not want to use the actual date as column headers in the attendance table because those dates will vary across each course's attendance sheet. Instead you should structure your table to use Day 1, Day 2, Day 3, etc. as column headers.
 - * You will also want to use row filtering to remove the "Name" row as well as empty rows.
4. Create a new parameter named FilePath using the path from the Source step for Attendance. Then click the gear icon for that step, and change the file path to your parameter.
5. Create a new function from AttendanceExample, and name it **AttendanceTransform**.
6. Now add a new data source to bring in the folder with all the attendance spreadsheets, and choose **Transform Data**.
 - * Why not **Combine and Transform**? What happens if you simply combine these spreadsheets?
7. Use row filtering to remove any files that were imported from this folder which you do not want to include in your analysis.
8. Remove all columns except for the **Name** and **Folder Path** columns.
 - * Wait, what was in that **Content** column? Did we need that? Nope, you will be bringing this back in a different way using the custom function you just created.
9. Merge the **Name** and **Folder Path** columns. Make sure **Folder Path** comes first in the full merged filepath.
10. Go to the **Add Column** tab, and select **Invoke Custom Function**. Name your new column **Content**.
11. Remove the **Folder Path** and **FullFilePath** columns. These have served their purpose. Keep the **Name** column, but rename it to **FileName**.

12. Click the button in the **Content** column header to expand the table data. Uncheck **Use original column name as prefix**.
13. Save your file as **MultiAttendance_Step01.pbix**

Transforming the Combined Data

1. Utilizing the above concepts, recreate the **Course** and **Students** dimension tables from the previous set of exercises.
 - * Hint: Think about which pieces you can reuse or duplicate to streamline the creation of additional tables.
 - * In the previous exercise, you manufactured a **CourseID** to use as a foreign key for the course. Is there a handy piece of data in the **Attendance** table you can use instead?
2. Reflect on the **Course Day** table from the previous set of exercises. Now that you have multiple course dates to contend with, can you structure a simpler dimension table that usefully ties together the day of course with the date? (Hint: Consider that these courses have overlapping dates, so you'll want to consider the common denominator across courses for tracking attendance. You might also want to create a new composite key to help tie your tables together.)
3. You may also choose to add a separate date table. There are multiple methods of creating a date table which you are welcome to investigate, but here's one quick way to do it:
 - * Identify the target date range you want to generate, e.g. full-year 2023.
 - * Create a blank query named **Date**. In the formula bar, create a list of dates for the year 2023 using the formula: `=List.Dates(#date(2023,1,1),365,#duration(1,0,0,0))`
 - * Convert the list to a table by clicking **To Table** in the **Transform** tab.
 - * Rename the column to **Date** and change the type to **Date**.
 - * Select the **Date** column. In the **Add Column** tab of the ribbon, use the **Date** drop-down options to add columns for Year, Month Number, Month Name, Day of Week, and Name of Day.
4. As a tidying-up step, organize your tables into groups for **Data Prep**, **Fact Tables**, and **Dimensions**.
5. Save your changes as **MultiAttendance_Step02.pbix**

Modeling & Visualizing the Combined Data

1. Create a star schema for your tables, and hide unused tables.
 - * Encounter any snags with “many to many” relationships? One student took the course twice, so their email address appears as a duplicate in the **Students** table when using **Email** as a foreign key.
 - ◇ What are some different ways you might handle this issue? Why might you choose one option versus another, from an analytics perspective?
 - ◇ For the purpose of this exercise, go ahead and return to Power Query to revisit the **Students** table, and right click on the **Email** column to remove duplicates. In a sneaky twist, there are differences in capitalization between the duplicate email addresses, so you will also need to format the column to lowercase before removing duplicates. (Hint: You’ll find this option in the Transform tab.)
 - * Remember to hide any fields that you will not be using directly in visualizations.
2. Now add measures and calculated columns, using the previous Attendance Analytics project for reference. Consider the different goals inherent in analyzing this larger data set. Are there other measures that would make sense here?
3. Start creating visualizations. Some ideas you might incorporate:
 - * How many total students have taken this course across all instances?
 - * What is the total attendance for all courses? How might you depict comparative attendance for each individual course?
 - * What attendance trends might you highlight?
4. Save your changes as **MultiAttendance_Step03.pbix**